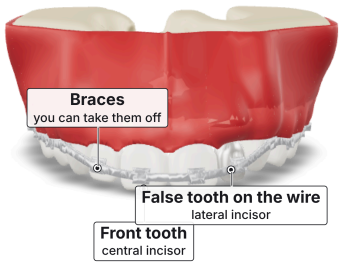
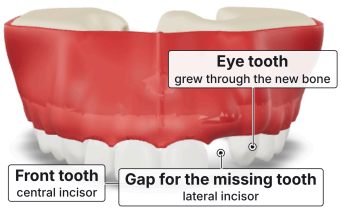
Stage 2: During braces

Bone graft for a cleft in the gum: the three stages

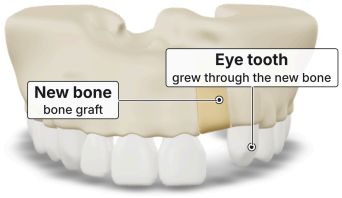
This model shows the same jaw about 2 to 3 years after the bone graft. The child is about 12 to 13 years old. The eye tooth has come down through the new bone. Fixed braces line up the teeth.



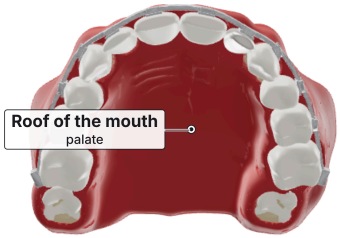
Whole jaw: braces on, with the false tooth



Braces taken off: the gap where the small tooth never grew



Gum taken off: the new bone and the eye tooth



From below: the roof of the mouth and the braces

What has changed

Bone now joins the two parts of the upper jaw where the gap was.[1,2]

The eye tooth came down through the graft into the row of teeth.[3,4]

Take off the pink gum, and you can see the new bone. It is the slightly yellow, rougher part next to the front tooth.

The braces

Fixed braces line up the teeth. Almost everyone who has this graft needs them.[1,2]

A white false tooth on the brace wire fills the gap where the small tooth never grew. It holds the space until the braces come off.

The pictures show the jaw with and without the braces.

Sources **1.** Bergland O, Semb G, Abyholm FE. Elimination of the residual alveolar cleft by secondary bone grafting and subsequent orthodontic treatment. Cleft Palate J. 1986;23:175-205. PMID 3524905. pubmed.ncbi.nlm.nih.gov/3524905 **2.** Semb G. Alveolar bone grafting. Front Oral Biol. 2012;16:124-136. doi.org/10.1159/000337666 **3.** Rivers J, et al. A clinical index scale for secondary alveolar bone graft success based on canine eruption. Eur J Orthod. 2025;47:cjae065. doi.org/10.1093/ejo/cjae065 **4.** Vellone V, et al. Canine eruption after secondary alveolar bone graft in unilateral cleft lip and palate patients. J Craniofac Surg. 2017;28:1206-1210. doi.org/10.1097/SCS.0000000000003659

This sheet is for a study at Aichi Gakuin University and Nagoya University. Version 0.2, 4 November 2026.